

PROFESSIONAL SUMMARY

- AI/ML Engineer (MS, Arizona State University) specializing in multi-agent architectures, RAG pipelines, and agentic LLM systems with LangChain, LangGraph, PyTorch, and Hugging Face. Skilled in LoRA fine-tuning, prompt engineering (CoT, few-shot), LLM-as-Judge evaluation, and production deployment via FastAPI, vector databases, Docker, and AWS. .

SKILLS

- LLM & AI Systems:** LLMs (GPT, Gemini, LLaMA, Qwen), Transformers (BERT), LangChain, LangGraph, RAG Pipelines, GraphRAG, Vector Databases (ChromaDB), Embeddings-Based Retrieval, Knowledge Graphs, Multi-Agent Architectures, Agentic Workflows, LLM Orchestration, Prompt Engineering , LLM-as-Judge Evaluation, Faithfulness Verification, Ollama
- Machine Learning & Deep Learning:** PyTorch, TensorFlow, Hugging Face (Transformers, Accelerate), scikit-learn, XGBoost, CUDA, Deep Learning, NLP, Multimodal AI, Computer Vision, Explainable AI (SHAP, LIME), Time-Series Modeling, Model Evaluation & Optimization
- Frameworks & Serving:** FastAPI, Flask, Streamlit, TableGPT, BLIP2, Gemma, Keras, NLTK, spaCy, NetworkX, Scrapy
- Programming & Databases:** Python, SQL, R, TypeScript, JavaScript, Java, C++, Shell/Bash, PostgreSQL, MySQL, MS SQL Server, MongoDB, DuckDB
- DevOps, Cloud & MLOps:** Docker, Kubernetes, Git, GitHub Actions, Azure DevOps, MLflow, Apache Airflow, Linux, AWS (S3, SageMaker, Redshift), Grafana, PyTorch Profiler, JIRA, Selenium, Cypress
- Data Engineering & Visualization:** ETL/ELT Pipelines, Data Warehousing, Spark, Hadoop, Parquet, dbt, Snowflake, Tableau, Power BI, Highcharts

EXPERIENCE

- Arizona State University – Decision Theater Network

Tempe, AZ

Data Analyst – Student Worker

May 2025 – Present

 - In-house Table Querying Chatbot:** Architected an LLM-powered chatbot integrating TableGPT with LangGraph-based agentic routing, ChromaDB for vector retrieval, and Streamlit for interactive visualization, deployed via FastAPI for real-time tabular insights.
 - Exploratory Data Analysis:** Conducted EDA and statistical validation using Python (Pandas, NumPy) to verify model outputs and ensure data quality across diverse datasets.
 - Visualization Development:** Designed interactive dashboards using Highcharts, Matplotlib, Seaborn, and Tableau; validated LLM-generated visualizations against business requirements and created summary reports in Excel.
 - Workflow Optimization:** Optimized TableGPT inference on Intel Gaudi accelerators using BF16/FP16 precision, quantization, and pipeline parallelism, achieving GPU-equivalent latency approximately 90% of the time for production-grade rapid responses.
- Hewlett Packard Enterprise (Aruba Networking)

Bangalore, India

Cloud Engineer

Jan. 2023 – Aug. 2024

 - Enhanced Aruba’s SD-Branch solution:** Engineered production features including traceroute and dynamic DNS for Aruba’s SD-Branch platform, improving network reliability by 30%.
 - Automated Deployment:** Automated scalable infrastructure provisioning and CI/CD workflows using Docker, Kubernetes, YANG models, and Azure DevOps on Linux-based environments.
 - Workflow Optimization:** Automated REST API and GraphQL scripting workflows, increasing engineering productivity by 50%.
 - Testing and Quality Assurance:** Reduced critical bugs by 95% through systematic manual and automated testing using JIRA, Selenium, and Cypress.
 - Agile Methodologies:** Streamlined Agile development processes across cross-functional teams, reducing project delivery time by 20%.
- Indegene Limited

Bangalore, India

Data Engineer Intern

Jun. 2022 – Aug. 2022

 - Data Migration:** Ensured 100% data integrity during migrations from SFTP, Azure, and SFMC platforms using SQL Server stored procedures, improving data reliability and system performance.
 - ETL and Big Data Processing:** Engineered data pipelines orchestrated with Apache Airflow using Python, Spark, and Hadoop, reducing processing time by 30% and enabling efficient stakeholder reporting.
 - Data Warehousing:** Optimized AWS S3 and Redshift workflows with PySpark, improving transfer efficiency by 25%; monitored pipeline health using Grafana dashboards.

EDUCATION

- Arizona State University

Arizona, AZ

Master of Science in Data Science, Analytics and Engineering; GPA: 4.0

Aug. 2024 – May. 2026
- PES University

Bangalore, India

Bachelor of Technology in Computer Science; GPA: 3.68 (9.06/10.0)

Aug. 2019 – May. 2023

PROJECTS

- **Multi-Agent Framework for Context-Aware Table Summarization:** A multi-agent LLM pipeline using LangChain and LangGraph to orchestrate specialized models across modalities TableGPT for tabular QA, Gemini for text QA, BLIP2 for image captioning, and Gemma for summary generation with LoRA fine-tuning via Hugging Face Accelerate. Applied Chain-of-Thought reasoning, instruction prompting, and faithfulness verification to improve output quality and reduce hallucinated summaries. Evaluated on TATQA and MMQA benchmarks using an LLM-as-Judge pipeline measuring answer relevance, grounding, and summary faithfulness.
- **PennyPath: AI Financial Planning Platform:** Developed an AI-driven financial planning platform using GraphRAG, a knowledge graph built from verified financial documents, using Gemini based personalization to deliver user-specific recommendations. Designed a hybrid pipeline combining persona-aware retrieval, deterministic rule-based prioritization, and interactive graph-backed explanations so users could see why plans were ordered a certain way. Improved responsiveness by structuring What-If scenarios to re-run only the Financial Order of Operations, cutting update latency to ~50ms and supporting real-time interaction through Flask APIs.
- **RateMyRep HackASU 2026 AI & Agents Engineer:** Built an agent-driven pipeline using Claude Code, Gemini 2.5 Flash, and TinyFish API to extract, classify, and reconcile legislator data from bills, voter guides, and public statements. Used Claude for PDF parsing and statement consolidation from Arizona Clean Elections voter guides, and designed prompting workflows for Gemini to perform batched topic classification across 12 policy categories. Integrated TinyFish as a web-scraping agent to gather public statements for 30 representatives, with PostgreSQL-backed storage and backend services deployed through InsForge.
- **AI/NLP 3D Interior Modeling:** Built a 3D interior modeling system by combining NLP and deep learning to translate user-generated text into room layouts and visual design elements. Used BERT for language understanding, Graph Convolutional Networks (GCNs) to model spatial relationships from textual inputs, and GANs for interior texture generation. Trained and tuned the pipeline in TensorFlow, leveraging GPU-based experimentation to optimize model performance and generation quality.
- **CMI BFRB Detection (Helios Wearable Multimodal Time-Series Classification):** Built a multimodal time-series classification pipeline to detect Body-Focused Repetitive Behaviors (BFRBs). Implemented sequence windowing (uniform timesteps), normalization/standardization, and missing-value handling for model-ready tensors. Trained a Conv1D baseline (BatchNorm/ReLU/MaxPool + Global Average Pooling) and achieved 96.02% Binary F1.
- **Hospital Price Transparency QA System (Ongoing):** Developed AI-powered question-answering system for healthcare price transparency using GraphRAG for procedure similarity search and knowledge retrieval across 120 Arizona hospitals and 97 MRF files. Integrated TableGPT for tabular question answering and DuckDB for SQL-based price analysis. Applied prompt engineering with Qwen LLM to enable natural language queries about procedure pricing, plan comparisons, and cross-hospital cost analysis. Automated data collection via Scrappy and structured 4GB of pricing data for AI-driven insights.
- **Threat Detection with Explainable AI:** Built a fraud detection pipeline for credit card transaction data using **XGBoost** and scikit-learn, improving prediction accuracy by 20% over a baseline model. Integrated **SHAP** and **LIME** to explain model decisions and identify influential features for interpretable fraud classification. Incorporated **TensorFlow**, **Hugging Face**, and transformer-based components for downstream explanation and analysis, strengthening the transparency and actionability of the detection workflow.

PUBLICATION

Kulkarni, R. P., & Puttane, S. M. (2024). Framework for Home Layout Design by Semantic Parsing of Text. *Advances in Computational Intelligence and Its Applications*, 195.